In [76]:
```
# This is an analysis of the mortgage dataset from kaggle
# https://www.kaggle.com/datasets/datawitharyan/financial-loan-dataset?re
```

In [1]:
```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
plt.style.use('ggplot')
pd.set_option('display.max_columns', 200)
from sklearn.model_selection import train_test_split
```

In [2]:
```
df = pd.read_excel(f"financial_loan.xlsx")
```

Generalities of the dataset

In [ ]:
```
df
```

Out[ ]:

	id	address_state	application_type	emp_length	emp_title	grade	hom
0	1077430	GA	INDIVIDUAL	< 1 year	Ryder	C	
1	1072053	CA	INDIVIDUAL	9 years	MKC Accounting	E	
2	1069243	CA	INDIVIDUAL	4 years	Chemat Technology Inc	C	
3	1041756	TX	INDIVIDUAL	< 1 year	barnes distribution	B	
4	1068350	IL	INDIVIDUAL	10+ years	J&J Steel Inc	A	
...	...	...	...	...	...	...	
38571	803452	NJ	INDIVIDUAL	< 1 year	Joseph M Sanzari Company	C	
38572	970377	NY	INDIVIDUAL	8 years	Swat Fame	C	
38573	875376	CA	INDIVIDUAL	5 years	Anaheim Regional Medical Center	D	
38574	972997	NY	INDIVIDUAL	5 years	Brooklyn Radiology	D	
38575	682952	NY	INDIVIDUAL	4 years	Allen Edmonds	F	

38576 rows × 24 columns

In [ ]:
```
df.columns
```

Out[ ]:
```
Index(['id', 'address_state', 'application_type', 'emp_length', 'emp_tit
le',
       'grade', 'home_ownership', 'issue_date', 'last_credit_pull_date',
       'last_payment_date', 'loan_status', 'next_payment_date', 'member_
id',
       'purpose', 'sub_grade', 'term', 'verification_status', 'annual_in
come',
       'dti', 'installment', 'int_rate', 'loan_amount', 'total_acc',
       'total_payment'],
      dtype='object')
```

In [ ]:
```
df.describe()
```

Out[ ]:

	id	issue_date	last_credit_pull_date	last_payment_date	nex
count	3.857600e+04	38576	38576	38576	
mean	6.810371e+05	2021-07-16 02:31:35.562007040	2021-06-08 13:36:34.193280512	2021-06-26 09:52:08.909166080	20
min	5.473400e+04	2021-01-01 00:00:00	2021-01-08 00:00:00	2021-01-08 00:00:00	20
25%	5.135170e+05	2021-04-11 00:00:00	2021-04-15 00:00:00	2021-03-16 00:00:00	20
50%	6.627280e+05	2021-07-11 00:00:00	2021-05-16 00:00:00	2021-06-14 00:00:00	20
75%	8.365060e+05	2021-10-11 00:00:00	2021-08-13 00:00:00	2021-09-15 00:00:00	20
max	1.077501e+06	2021-12-12 00:00:00	2022-01-20 00:00:00	2021-12-15 00:00:00	20
std	2.113246e+05	NaN	NaN	NaN	

Small data cleanup

Assign right datatypes to the features

In [ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 38576 entries, 0 to 38575
Data columns (total 24 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   id                     38576 non-null  int64
 1   address_state          38576 non-null  object
 2   application_type        38576 non-null  object
 3   emp_length             38576 non-null  object
 4   emp_title              37138 non-null  object
 5   grade                  38576 non-null  object
 6   home_ownership         38576 non-null  object
 7   issue_date             38576 non-null  datetime64[ns]
 8   last_credit_pull_date  38576 non-null  datetime64[ns]
 9   last_payment_date      38576 non-null  datetime64[ns]
 10  loan_status            38576 non-null  object
 11  next_payment_date      38576 non-null  datetime64[ns]
 12  member_id              38576 non-null  int64
 13  purpose                38576 non-null  object
 14  sub_grade              38576 non-null  object
 15  term                   38576 non-null  object
 16  verification_status    38576 non-null  object
 17  annual_income          38576 non-null  float64
 18  dti                    38576 non-null  float64
 19  installment            38576 non-null  float64
 20  int_rate               38576 non-null  float64
 21  loan_amount            38576 non-null  int64
 22  total_acc              38576 non-null  int64
 23  total_payment          38576 non-null  int64
dtypes: datetime64[ns](4), float64(4), int64(5), object(11)
memory usage: 7.1+ MB
```

In [ ]:
```
# Deal with categorical data

cat_cols = [
    'address_state',
    'application_type',
    'grade',
    'loan_status',
    'purpose',
    'sub_grade',
    'term',
    'verification_status',
    'home_ownership'
]

df[cat_cols] = df[cat_cols].astype('category').copy()
```

In [ ]:
```
df.term.unique()
```

Out[ ]:
```
[' 60 months', ' 36 months']
Categories (2, object): [' 36 months', ' 60 months']
```

In [ ]:
```
# Deal with employment length

def clean_emp_length(x):
    if pd.isna(x):
        return np.nan
    x = str(x)
    x = x.strip()
```

```
    if x == '< 1 year':
        return 0
    if x == '10+ years':
        return 10
    return int(x.split()[0])

df['emp_length'] = df['emp_length'].apply(clean_emp_length).copy()
```

In [ ]:
```
# Deal with mortgage grade

grade_order = ['A', 'B', 'C', 'D', 'E', 'F', 'G']
df['grade'] = pd.Categorical(df['grade'], categories=grade_order, ordered




sub_grade_order = list(df['sub_grade'].unique().sort_values())
df['sub_grade'] = pd.Categorical(df['sub_grade'], categories=sub_grade_or
```

In [ ]:
```
# Deal with datetime
date_cols = [
    "issue_date",
    "last_credit_pull_date",
    "last_payment_date",
    "next_payment_date",
]

df[date_cols] = df[date_cols].apply(pd.to_datetime)
```

In [ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 38576 entries, 0 to 38575
Data columns (total 24 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   id                     38576 non-null  int64
 1   address_state          38576 non-null  category
 2   application_type        38576 non-null  category
 3   emp_length             38576 non-null  int64
 4   emp_title              37138 non-null  object
 5   grade                  38576 non-null  category
 6   home_ownership         38576 non-null  object
 7   issue_date             38576 non-null  datetime64[ns]
 8   last_credit_pull_date  38576 non-null  datetime64[ns]
 9   last_payment_date      38576 non-null  datetime64[ns]
 10  loan_status            38576 non-null  category
 11  next_payment_date      38576 non-null  datetime64[ns]
 12  member_id              38576 non-null  int64
 13  purpose                38576 non-null  category
 14  sub_grade              38576 non-null  category
 15  term                   38576 non-null  category
 16  verification_status    38576 non-null  category
 17  annual_income          38576 non-null  float64
 18  dti                    38576 non-null  float64
 19  installment            38576 non-null  float64
 20  int_rate               38576 non-null  float64
 21  loan_amount            38576 non-null  int64
 22  total_acc              38576 non-null  int64
 23  total_payment          38576 non-null  int64
dtypes: category(8), datetime64[ns](4), float64(4), int64(6), object(2)
memory usage: 5.0+ MB
```

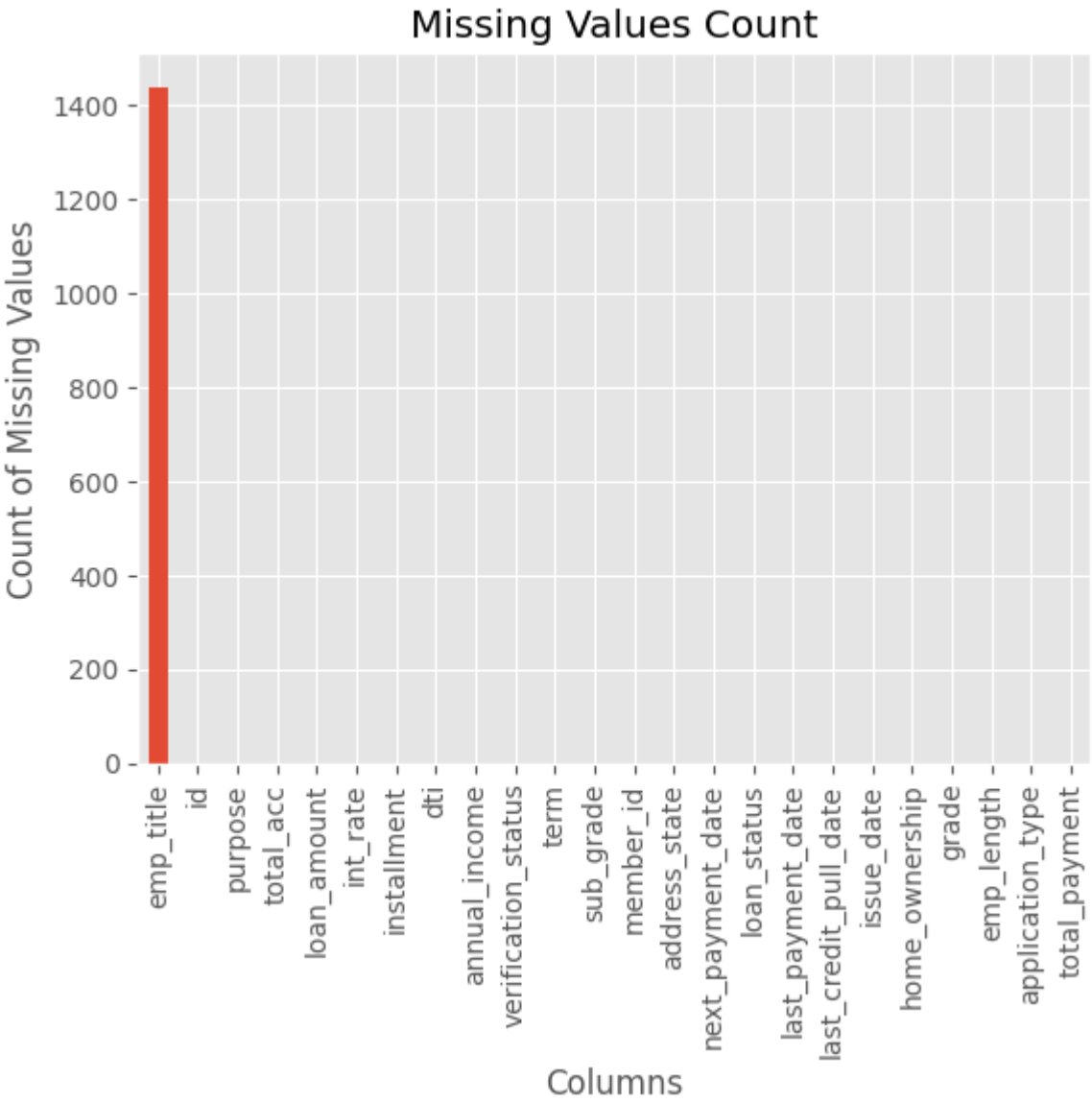
Find missing values

In [ ]:

```
## Check for missing values
print(df.isna().sum())
ax = df.isna().sum().sort_values(ascending=False).plot(kind='bar')

ax.set_title('Missing Values Count')
ax.set_xlabel('Columns')
ax.set_ylabel('Count of Missing Values')
```

```
id                          0
address_state               0
application_type            0
emp_length                  0
emp_title                1438
grade                       0
home_ownership              0
issue_date                  0
last_credit_pull_date       0
last_payment_date           0
loan_status                 0
next_payment_date           0
member_id                   0
purpose                     0
sub_grade                   0
term                        0
verification_status         0
annual_income               0
dti                         0
installment                 0
int_rate                    0
loan_amount                 0
total_acc                   0
total_payment               0
dtype: int64
```

Out[ ]: Text(0, 0.5, 'Count of Missing Values')



Look for duplicates

In [ ]:
```
## Check for duplicates
print(f"Number of fully duplicate rows: {df.duplicated().sum()}")

print("Shows the second occurrence of duplicate rows based on the 'emp_ti
df.loc[df.duplicated(subset = ['emp_title'], keep = 'last')]
```

```
Number of fully duplicate rows: 0
Shows the second occurrence of duplicate rows based on the 'emp_title' col
umn:
```

Out[ ]:

	id	address_state	application_type	emp_length	emp_title	grade	hom
6	1067441	TX	INDIVIDUAL	10	American Airlines	C	
10	1057766	TX	INDIVIDUAL	10	Ericsson	B	
15	1060995	RI	INDIVIDUAL	0	HSA-UWC	B	
19	1058060	MD	INDIVIDUAL	10	NaN	D	
21	207910	FL	INDIVIDUAL	0	NaN	A	
...	...	...	...	...	...	...	
38326	781758	CA	INDIVIDUAL	5	NaN	C	
38357	788049	NJ	INDIVIDUAL	5	AT&T	D	
38371	1017940	CA	INDIVIDUAL	10	California Department of Corrections	C	
38395	738177	NC	INDIVIDUAL	8	Travelers	D	
38498	689089	NY	INDIVIDUAL	10	us postal service	B	

10050 rows × 24 columns

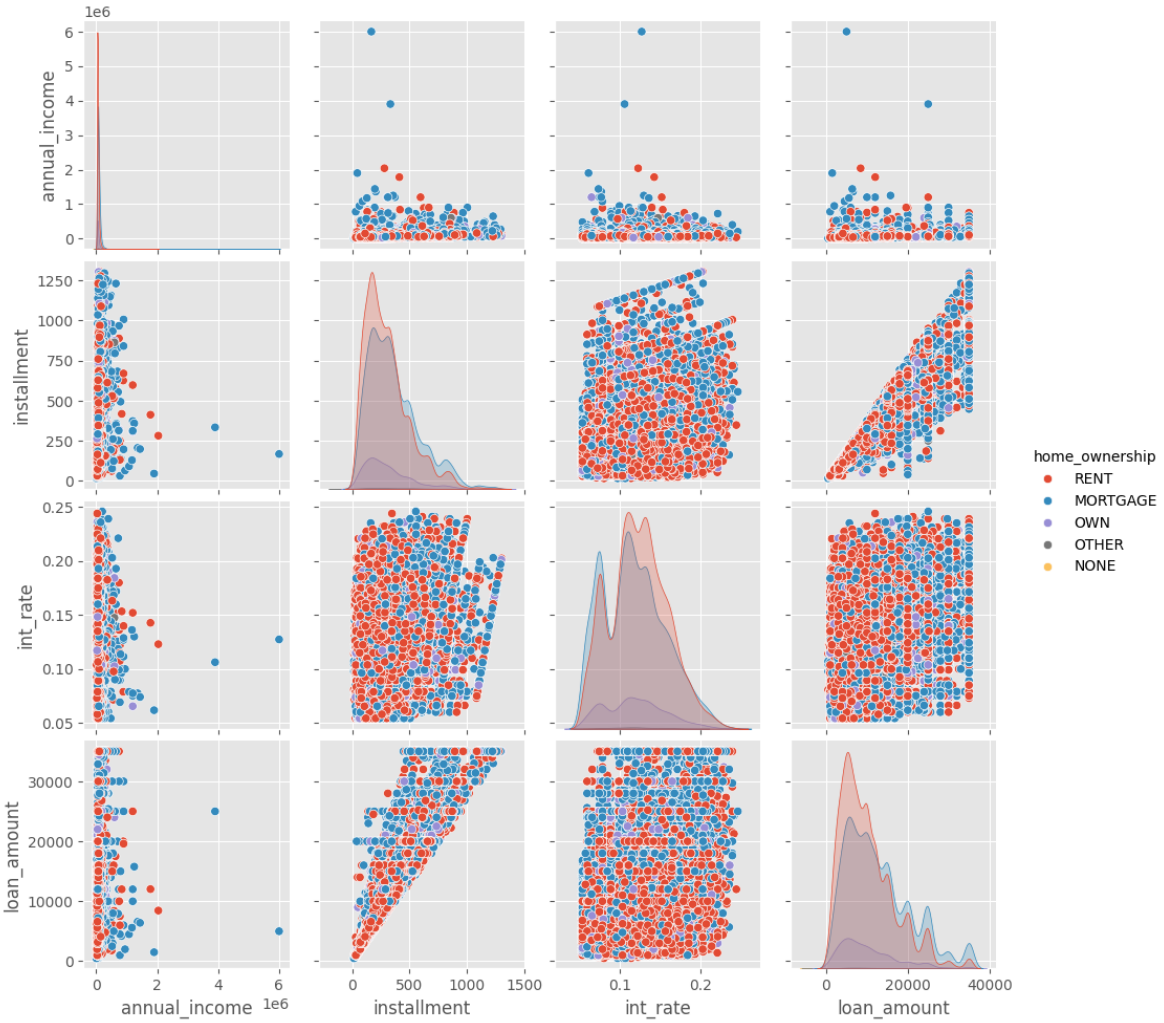
Feature Exploration

Feature understanding and visualization

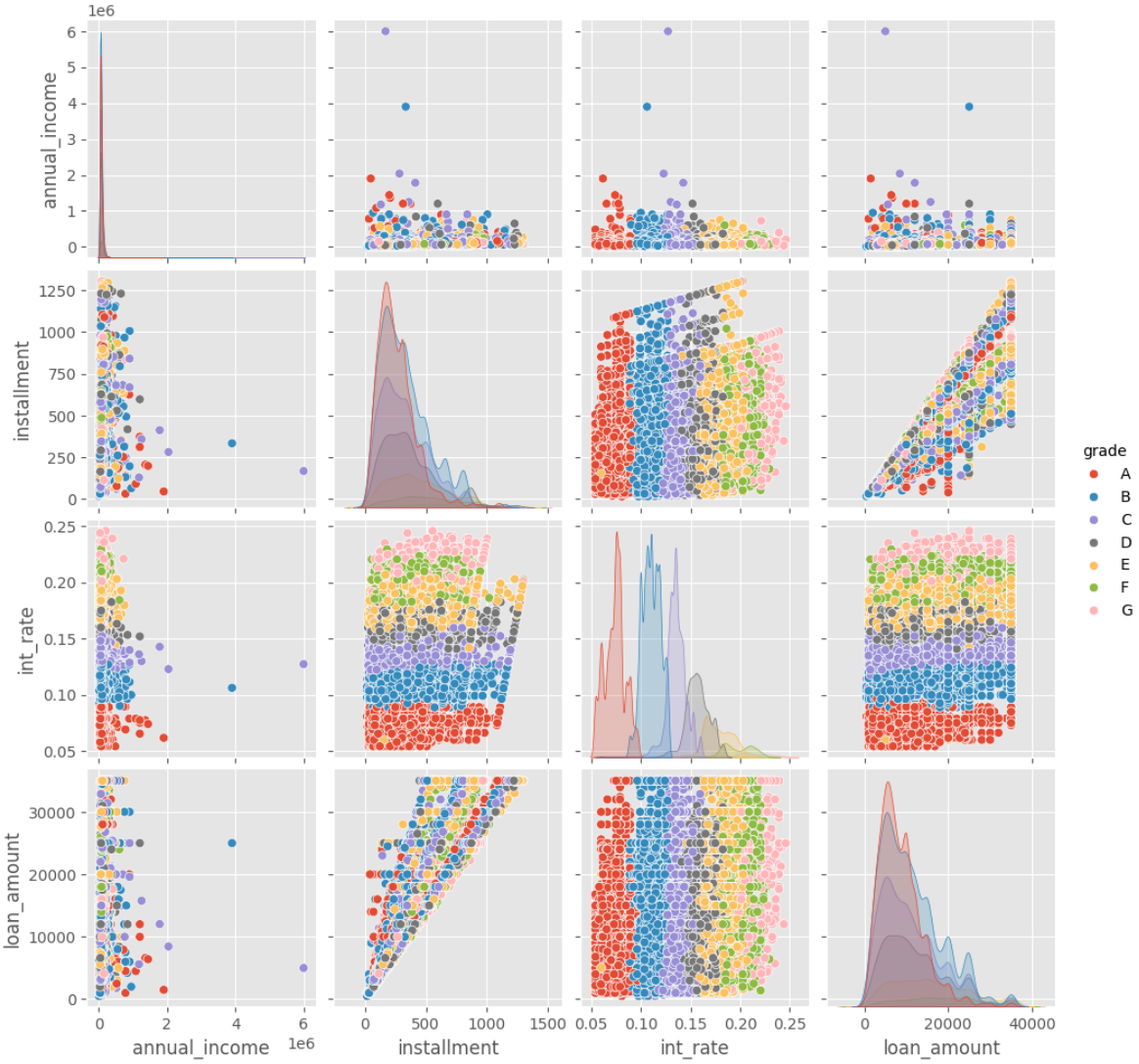
In [ ]:
```
df.columns
```

Out[ ]:
```
Index(['id', 'address_state', 'application_type', 'emp_length', 'emp_tit
le',
       'grade', 'home_ownership', 'issue_date', 'last_credit_pull_date',
       'last_payment_date', 'loan_status', 'next_payment_date', 'member_
id',
       'purpose', 'sub_grade', 'term', 'verification_status', 'annual_in
come',
       'dti', 'installment', 'int_rate', 'loan_amount', 'total_acc',
       'total_payment'],
      dtype='object')
```

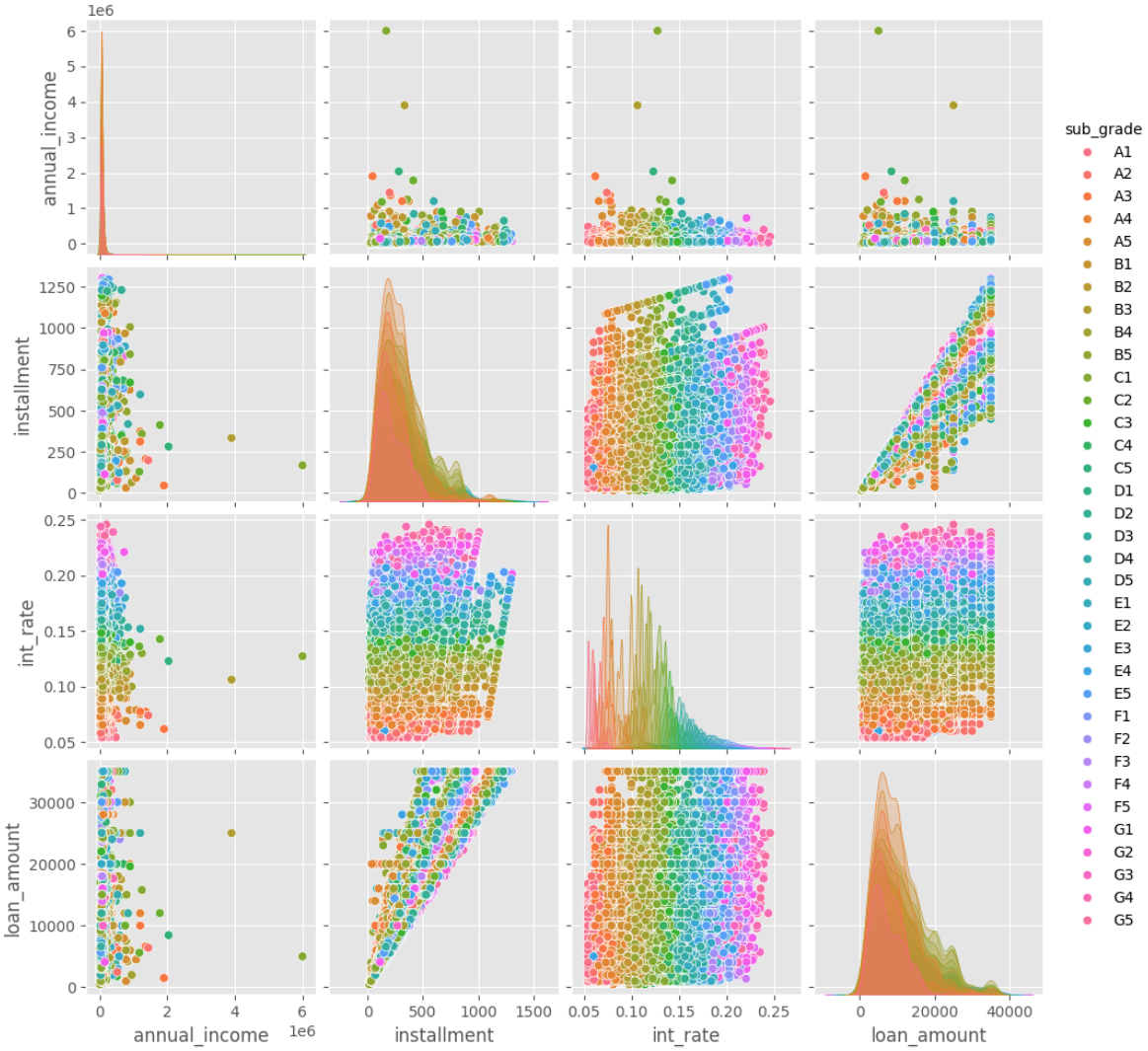
In [ ]:
```
sns.pairplot(df,
             vars=['annual_income',
                   'installment','int_rate','loan_amount'],
            hue='home_ownership')
plt.show()
```



In [ ]:
```
sns.pairplot(df,
             vars=['annual_income',
                   'installment','int_rate','loan_amount'],
            hue='grade')
plt.show()
```



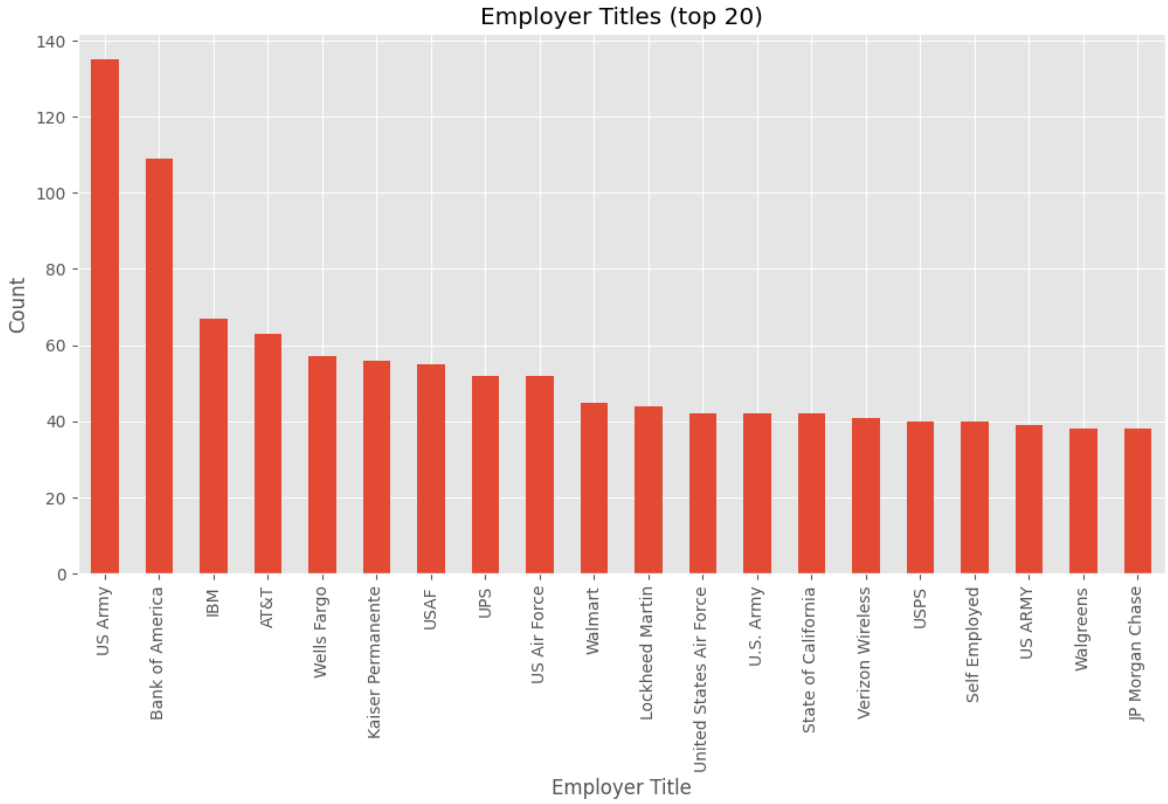
In [ ]:
```
sns.pairplot(df,
             vars=['annual_income',
                   'installment','int_rate','loan_amount'],
            hue='sub_grade')
plt.show()
```



In [ ]:

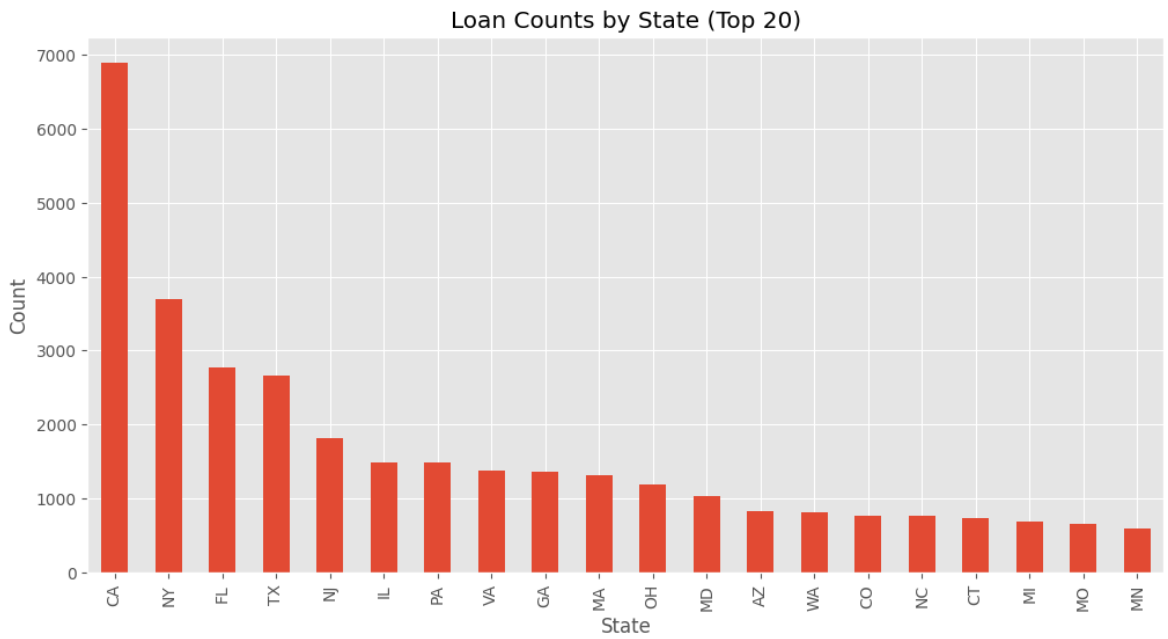
In [ ]:
```
ax = df.groupby('emp_title').size().sort_values(ascending=False).head(20)
ax.set_title('Employer Titles (top 20)')
ax.set_xlabel('Employer Title')
ax.set_ylabel('Count')
```

Out[ ]: Text(0, 0.5, 'Count')



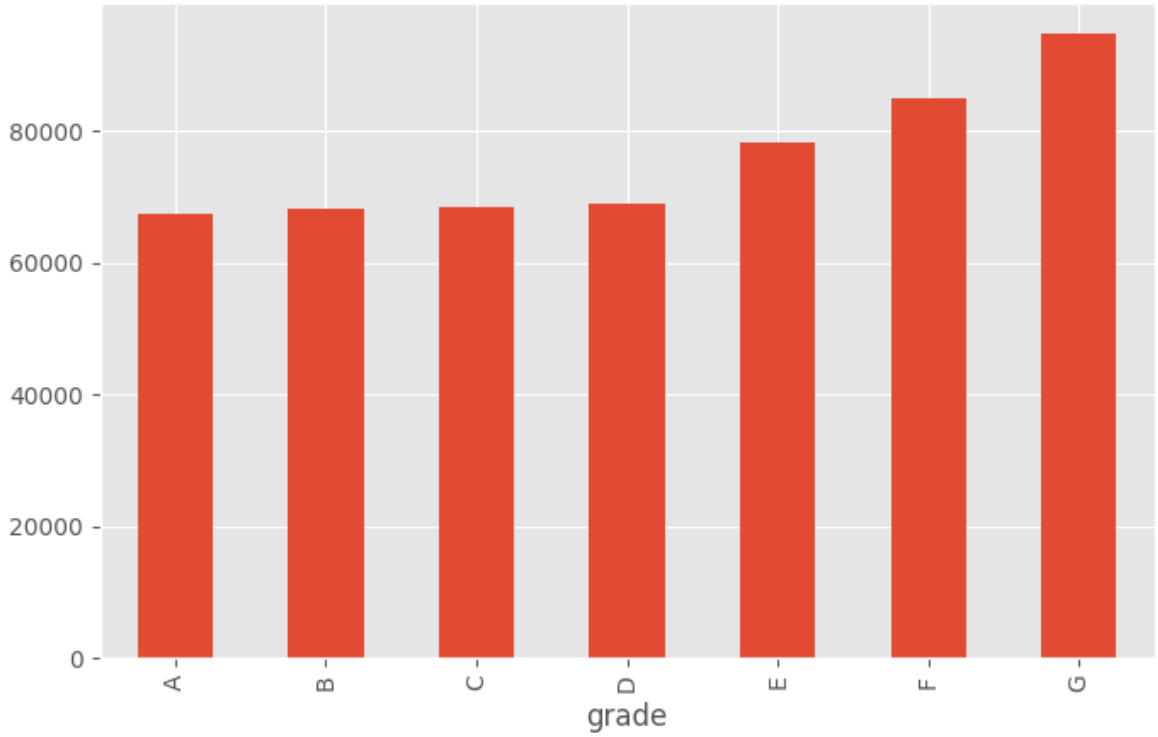
In [ ]:
```
df.groupby('address_state').size().sort_values(ascending=False).head(20).
plt.title('Loan Counts by State (Top 20)')
plt.xlabel('State')
plt.ylabel('Count')
```

Out[ ]: Text(0, 0.5, 'Count')



In [ ]:
```
df.groupby('grade')["annual_income"].mean().plot(kind='bar', figsize=(8,
```

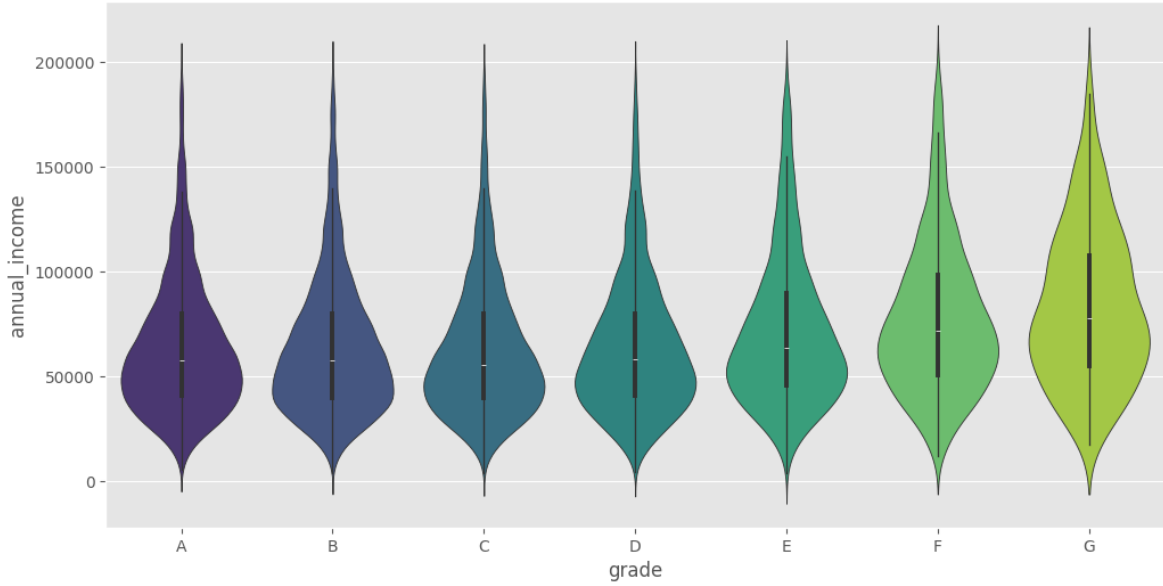
Out[ ]: <Axes: xlabel='grade'>



In [ ]:
```
plt.figure(figsize=(12, 6))
sns.violinplot(x='grade', y='annual_income', data=df[df['annual_income']
                order=['A','B','C','D','E','F','G'], palette='viridis')
```

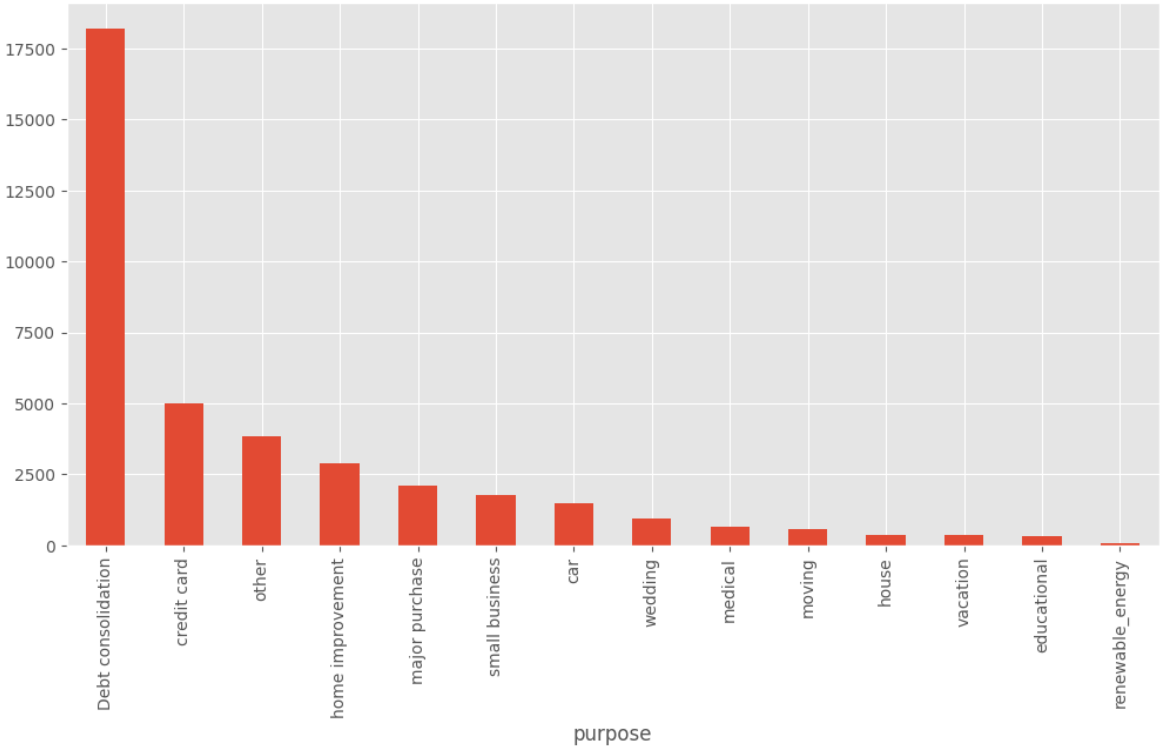
```
/tmp/ipykernel_67783/953955367.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be remove
d in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for
the same effect.

  sns.violinplot(x='grade', y='annual_income', data=df[df['annual_income']
< 200000],
```

Out[ ]:
```
<Axes: xlabel='grade', ylabel='annual_income'>
```



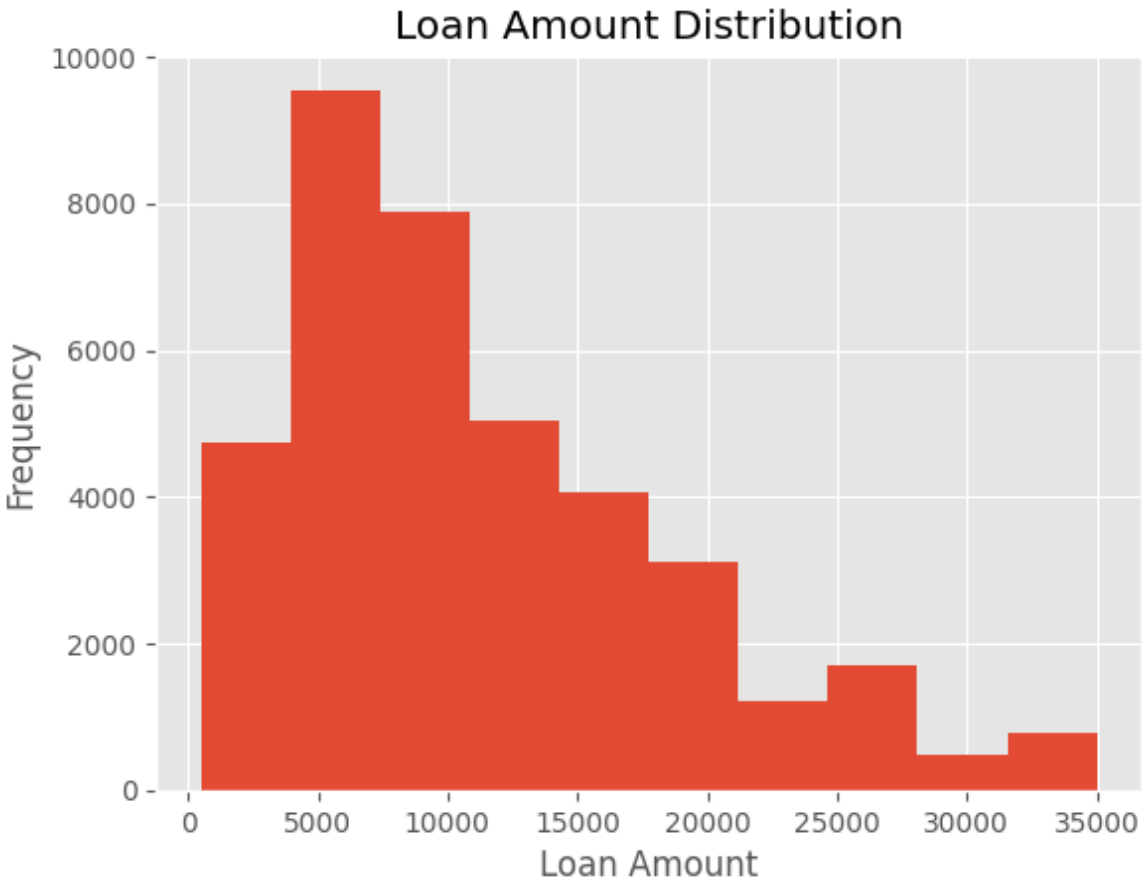
In [ ]:
```
df.purpose.value_counts().plot(kind='bar', figsize=(12, 6))
```

Out[ ]:
```
<Axes: xlabel='purpose'>
```



In [ ]:
```
ax = df['loan_amount'].plot(kind='hist',
                            title='Loan Amount Distribution')
ax.set_xlabel('Loan Amount')
```

Out[ ]: Text(0.5, 0, 'Loan Amount')



In [ ]:
```
df.columns
```

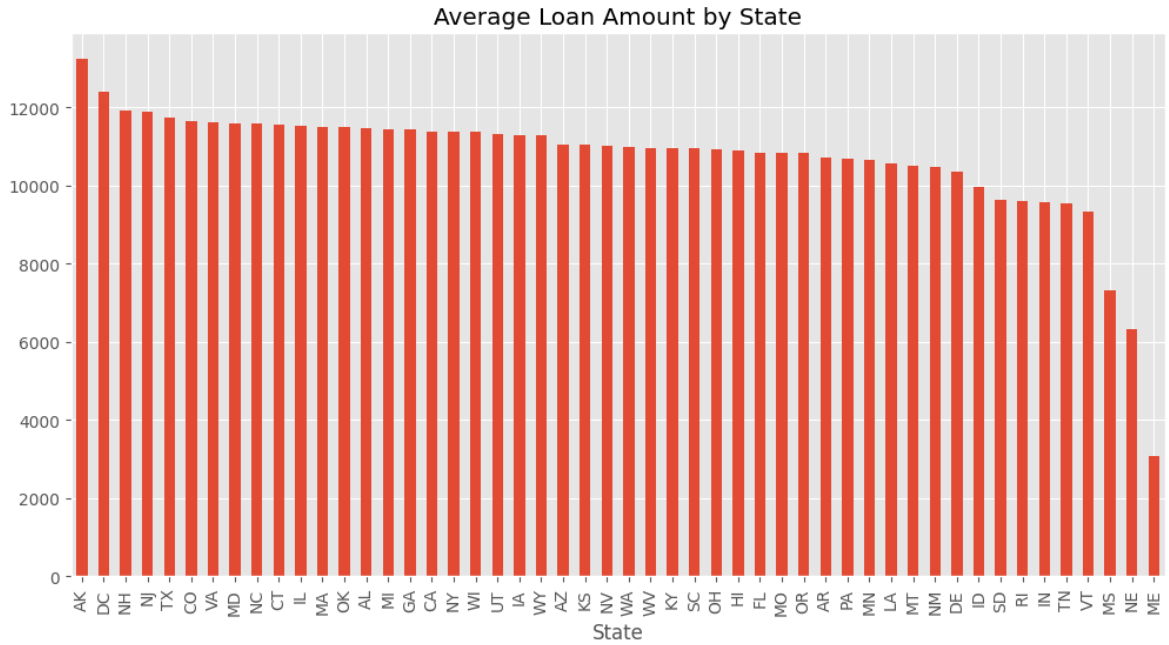
Out[ ]:
```
Index(['id', 'address_state', 'application_type', 'emp_length', 'emp_tit
le',
       'grade', 'home_ownership', 'issue_date', 'last_credit_pull_date',
       'last_payment_date', 'loan_status', 'next_payment_date', 'member_
id',
       'purpose', 'sub_grade', 'term', 'verification_status', 'annual_in
come',
       'dti', 'installment', 'int_rate', 'loan_amount', 'total_acc',
       'total_payment'],
      dtype='object')
```

Practice questions

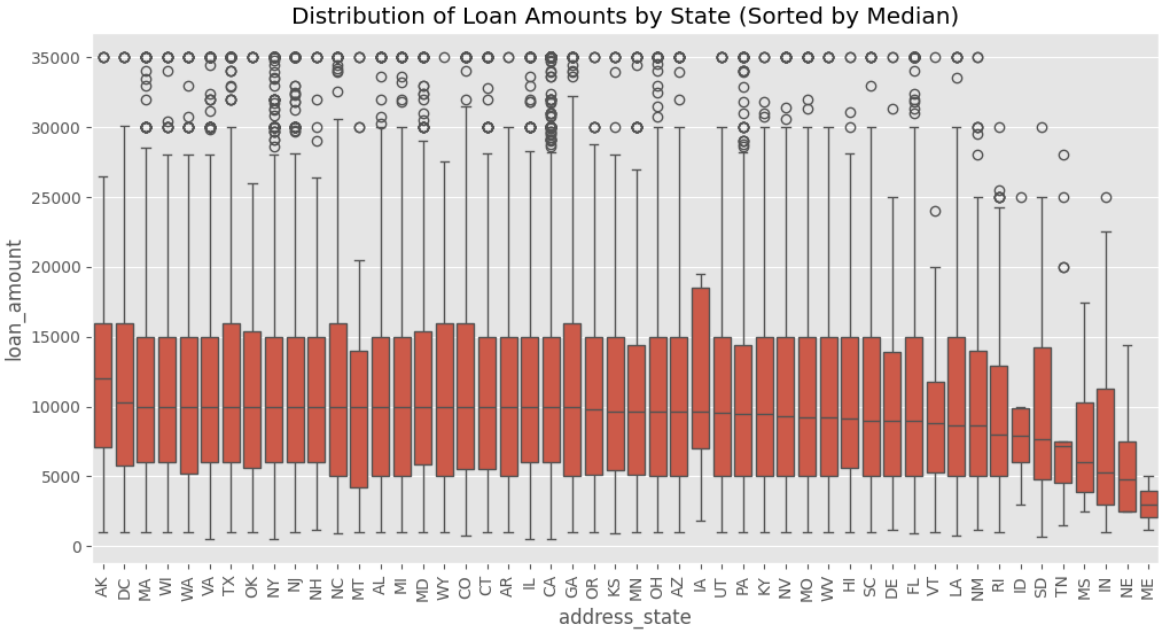
- What state has the largest loans?

In [ ]:
```
ax = df.groupby("address_state")["loan_amount"].mean().sort_values(ascend


ax.set_title("Average Loan Amount by State")
ax.set_xlabel("State")
```

Out[ ]:
```
Text(0.5, 0, 'State')
```



In [ ]:
```
fig,ax = plt.subplots(figsize=(12,6))

state_order = df.groupby("address_state")["loan_amount"].median().sort_va

sns.boxplot(x=df["address_state"], y=df["loan_amount"],ax=ax, order=state

plt.xticks(rotation=90)
ax.set_title("Distribution of Loan Amounts by State (Sorted by Median)")
plt.show()
```



In [ ]:

- Question: Make a model to predict the (sub-)grade of the mortgage

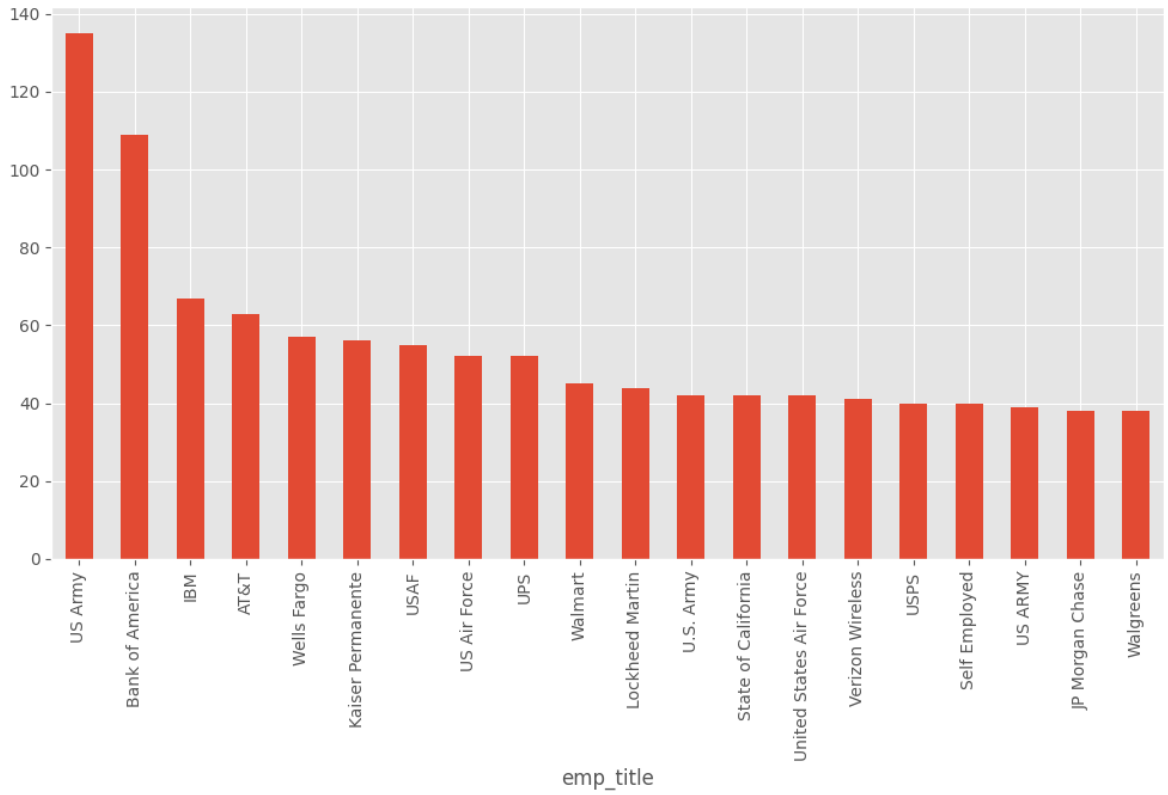
Feature engineering

Feature selection

Goal is to find the relevant features and which ones we can drop

In [ ]:
```
df['emp_title'].value_counts().head(20).plot(kind='bar', figsize=(12, 6))
print(len(df['emp_title'].unique()))
```

28526



In [ ]:
```
rdf = df.drop(columns=['id','member_id','issue_date','last_credit_pull_da
                       'next_payment_date', 'last_payment_date',
                       'application_type','grade',"emp_title"]).copy()

rdf
```

Out[ ]:

	address_state	emp_length	home_ownership	loan_status	purpose	sub_grade
0	GA	0	RENT	Charged Off	car	C4
1	CA	9	RENT	Fully Paid	car	E1
2	CA	4	RENT	Charged Off	car	C5
3	TX	0	MORTGAGE	Fully Paid	car	B2
4	IL	10	MORTGAGE	Fully Paid	car	A1
...	...	...	...	...	...	...
38571	NJ	0	MORTGAGE	Current	wedding	C1
38572	NY	8	RENT	Current	wedding	C1
38573	CA	5	RENT	Current	wedding	D5
38574	NY	5	RENT	Current	wedding	D5
38575	NY	4	RENT	Current	wedding	F3

38576 rows × 15 columns

Predictive Modeling

Make the train test split

In [ ]:

```
columns = [c for c in rdf.columns if c not in ['sub_grade']]

X = rdf[columns]
y = rdf['sub_grade']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,

X_train
```

Out[ ]:

	address_state	emp_length	home_ownership	loan_status	purpose	term
7542	WA	6	RENT	Charged Off	Debt consolidation	36 months
30069	NJ	6	RENT	Charged Off	major purchase	60 months
28709	NJ	10	OWN	Fully Paid	major purchase	36 months
4453	NY	1	RENT	Fully Paid	credit card	36 months
12655	NY	4	RENT	Fully Paid	Debt consolidation	36 months
...	...	...	...	...	...	...
4264	CA	4	RENT	Fully Paid	credit card	36 months
36107	AZ	1	MORTGAGE	Fully Paid	small business	36 months
5910	TX	6	RENT	Fully Paid	credit card	60 months
19268	NY	10	RENT	Fully Paid	Debt consolidation	36 months
35396	MD	0	MORTGAGE	Current	other	60 months

30860 rows × 14 columns

LightGBM

In [ ]:

```
import lightgbm as lgb
from sklearn.metrics import classification_report, accuracy_score, confus
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Initialize the LightGBM Classifier
# 'multi_logloss' is used for multiclass; 'binary_logloss' for two classe
model = lgb.LGBMClassifier(
    objective='multiclass', # Use 'binary' if you only have two categorie
    num_leaves=31,
    learning_rate=0.05,
    n_estimators=100,
    random_state=42,
    lambda_l1=1.5,  # Higher values increase sparsity
    lambda_l2=1.0,
)

# 2. Train the model
model.fit(
    X_train, y_train,
    eval_set=[(X_test, y_test)],
    eval_metric='multi_logloss',
    callbacks=[lgb.early_stopping(stopping_rounds=10)],
```

```
)

# 3. Make predictions
y_pred = model.predict(X_test)
```

```
[LightGBM] [Warning] lambda_l2 is set=1.0, reg_lambda=0.0 will be ignored.
Current value: lambda_l2=1.0
[LightGBM] [Warning] lambda_l1 is set=1.5, reg_alpha=0.0 will be ignored.
Current value: lambda_l1=1.5
[LightGBM] [Warning] lambda_l2 is set=1.0, reg_lambda=0.0 will be ignored.
Current value: lambda_l2=1.0
[LightGBM] [Warning] lambda_l1 is set=1.5, reg_alpha=0.0 will be ignored.
Current value: lambda_l1=1.5
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000910 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 1654
[LightGBM] [Info] Number of data points in the train set: 30860, number of
used features: 14
[LightGBM] [Warning] lambda_l2 is set=1.0, reg_lambda=0.0 will be ignored.
Current value: lambda_l2=1.0
[LightGBM] [Warning] lambda_l1 is set=1.5, reg_alpha=0.0 will be ignored.
Current value: lambda_l1=1.5
[LightGBM] [Info] Start training from score -3.601436
[LightGBM] [Info] Start training from score -3.287961
[LightGBM] [Info] Start training from score -3.098719
[LightGBM] [Info] Start training from score -2.622093
[LightGBM] [Info] Start training from score -2.676631
[LightGBM] [Info] Start training from score -3.092275
[LightGBM] [Info] Start training from score -2.964470
[LightGBM] [Info] Start training from score -2.611003
[LightGBM] [Info] Start training from score -2.754478
[LightGBM] [Info] Start training from score -2.680406
[LightGBM] [Info] Start training from score -2.916039
[LightGBM] [Info] Start training from score -2.973303
[LightGBM] [Info] Start training from score -3.253828
[LightGBM] [Info] Start training from score -3.468202
[LightGBM] [Info] Start training from score -3.511756
[LightGBM] [Info] Start training from score -3.744172
[LightGBM] [Info] Start training from score -3.379719
[LightGBM] [Info] Start training from score -3.518292
[LightGBM] [Info] Start training from score -3.693426
[LightGBM] [Info] Start training from score -3.813654
[LightGBM] [Info] Start training from score -3.940286
[LightGBM] [Info] Start training from score -4.098892
[LightGBM] [Info] Start training from score -4.273431
[LightGBM] [Info] Start training from score -4.456683
[LightGBM] [Info] Start training from score -4.544203
[LightGBM] [Info] Start training from score -4.776534
[LightGBM] [Info] Start training from score -5.069358
[LightGBM] [Info] Start training from score -5.353610
[LightGBM] [Info] Start training from score -5.469682
[LightGBM] [Info] Start training from score -5.815428
[LightGBM] [Info] Start training from score -5.942767
[LightGBM] [Info] Start training from score -6.210082
[LightGBM] [Info] Start training from score -6.673654
[LightGBM] [Info] Start training from score -6.530554
[LightGBM] [Info] Start training from score -7.159162
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
```

```
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
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[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
Did not meet early stopping. Best iteration is:
[100]	valid_0's multi_logloss: 0.252095
[LightGBM] [Warning] lambda_l2 is set=1.0, reg_lambda=0.0 will be ignored.
Current value: lambda_l2=1.0
[LightGBM] [Warning] lambda_l1 is set=1.5, reg_alpha=0.0 will be ignored.
Current value: lambda_l1=1.5
```
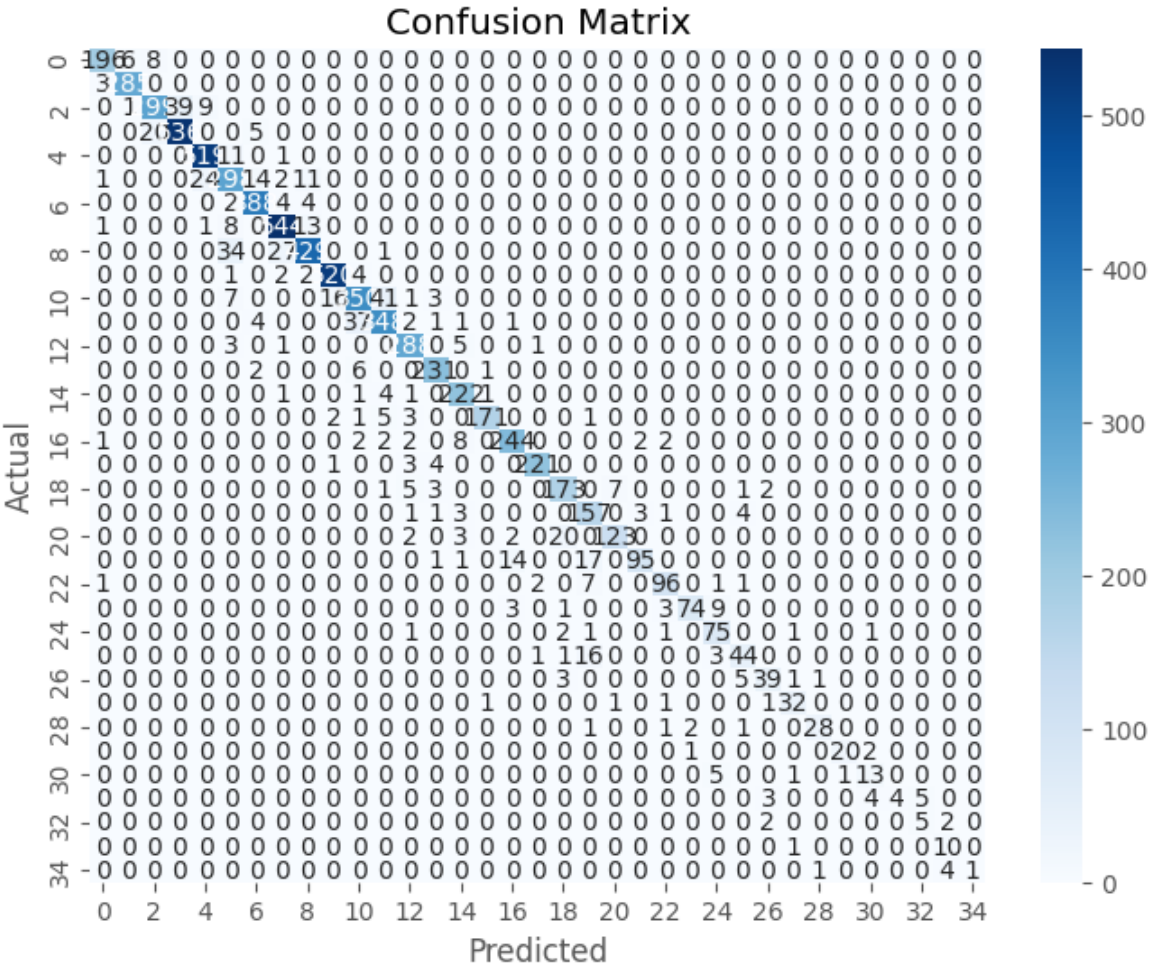
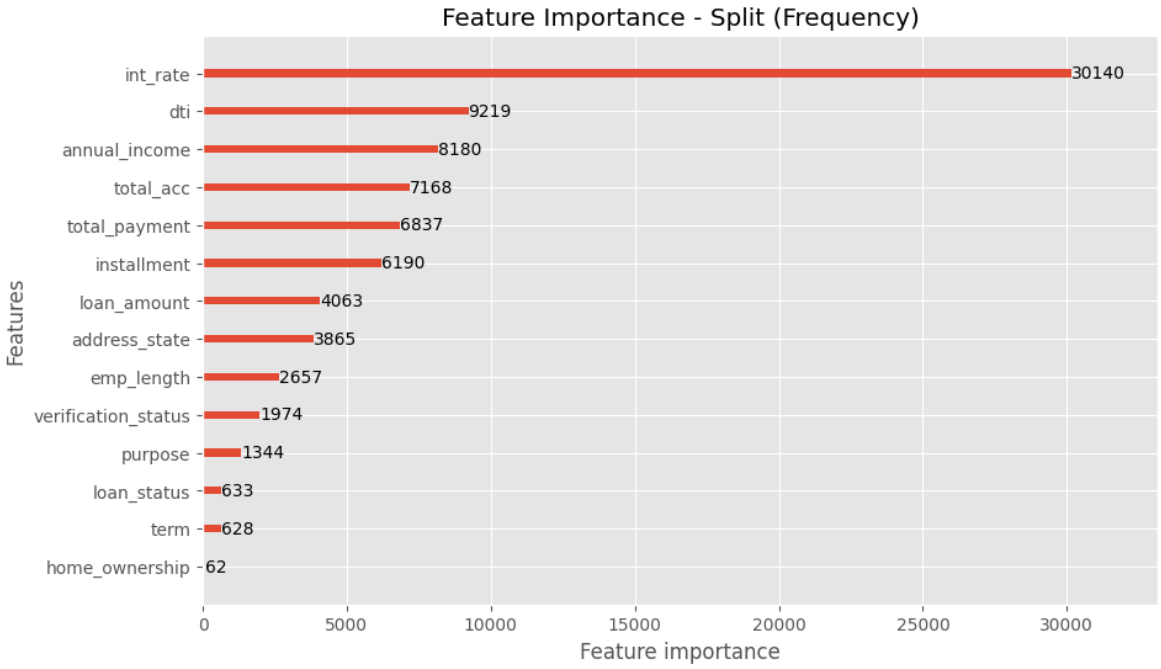

In [ ]:

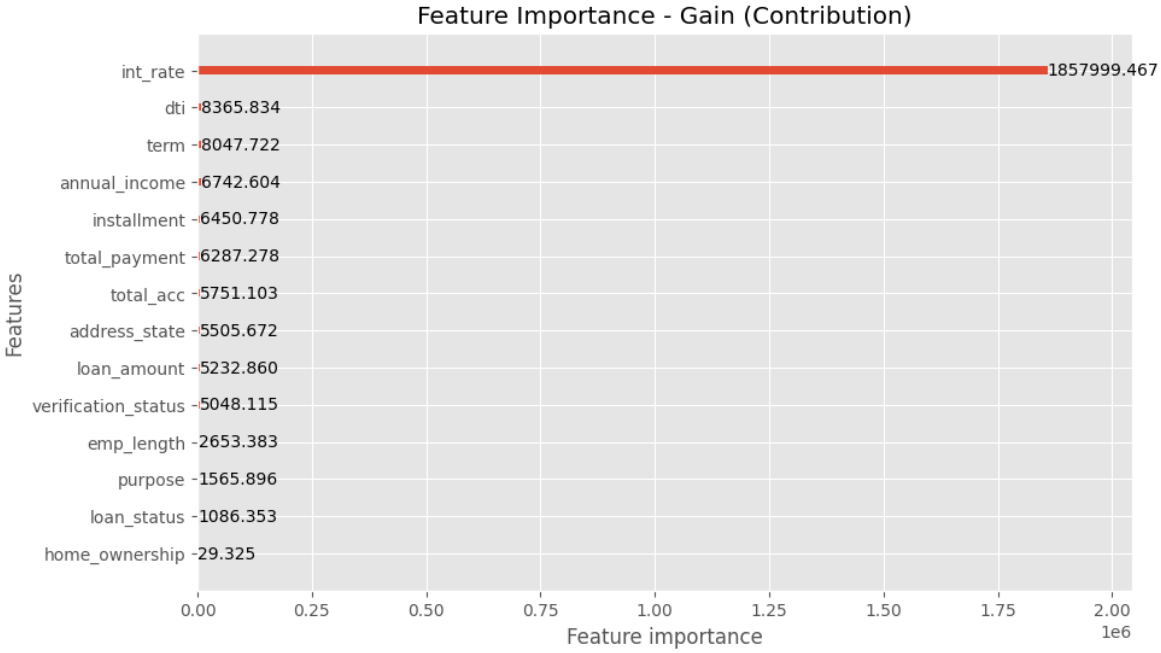
```python
# 4. Evaluation
print("--- Model Performance ---")
print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# 5. Visualizing the Confusion Matrix
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```

```
--- Model Performance ---
Accuracy: 0.9173

Classification Report:
              precision    recall  f1-score   support

          A1       0.97      0.93      0.95       210
          A2       0.98      0.99      0.98       288
          A3       0.91      0.86      0.89       348
          A4       0.93      0.96      0.94       561
          A5       0.94      0.98      0.96       531
          B1       0.82      0.85      0.83       350
          B2       0.94      0.97      0.96       398
          B3       0.93      0.96      0.95       567
          B4       0.93      0.87      0.90       491
          B5       0.96      0.98      0.97       529
          C1       0.87      0.84      0.85       418
          C2       0.87      0.88      0.87       394
          C3       0.93      0.97      0.95       298
          C4       0.95      0.96      0.95       240
          C5       0.91      0.97      0.94       230
          D1       0.98      0.93      0.96       183
          D2       0.92      0.93      0.93       263
          D3       0.98      0.97      0.97       229
          D4       0.86      0.90      0.88       192
          D5       0.79      0.92      0.85       170
          E1       0.94      0.82      0.88       150
          E2       0.95      0.74      0.83       128
          E3       0.91      0.89      0.90       108
          E4       0.96      0.82      0.89        90
          E5       0.81      0.91      0.86        82
          F1       0.79      0.68      0.73        65
          F2       0.83      0.80      0.81        49
          F3       0.89      0.89      0.89        36
          F4       0.93      0.85      0.89        33
          F5       0.95      0.87      0.91        23
          G1       0.65      0.65      0.65        20
          G2       1.00      0.25      0.40        16
          G3       0.50      0.56      0.53         9
          G4       0.62      0.91      0.74        11
          G5       1.00      0.17      0.29         6

    accuracy                           0.92      7716
   macro avg       0.89      0.84      0.85      7716
weighted avg       0.92      0.92      0.92      7716
```



In [ ]:
```
# 1. Importance by 'split' (Frequency)
# This shows how many times a feature was used to make a decision
lgb.plot_importance(model, importance_type='split', max_num_features=20,
plt.title("Feature Importance - Split (Frequency)")
plt.show()

# 2. Importance by 'gain' (Total Information Gain)
# This shows which features actually contributed the most to the model's
lgb.plot_importance(model, importance_type='gain', max_num_features=20, f
plt.title("Feature Importance - Gain (Contribution)")
plt.show()
```





LightGBM with cross validation

In [81]:
```
from sklearn.model_selection import RandomizedSearchCV

# 1. Initialize the base model
lgbm = lgb.LGBMClassifier(
    objective='multiclass',
    random_state=42,
    verbosity=-1 # Silences unnecessary warnings during search
)

# 2. Define the parameter grid
# We provide ranges for the model to "explore"
param_dist = {
    'n_estimators': [50, 100, 200],
    'learning_rate': [0.01, 0.05, 0.1],
    'num_leaves': [20, 31, 40, 60],
    'max_depth': [-1, 10, 20], # -1 means no limit
```

```
    'lambda_l1': [0, 1, 5],
    'lambda_l2': [0, 1, 5],
    'min_child_samples': [10, 20, 30]
}

# 3. Set up Randomized Search
# cv=5 means 5-fold cross-validation
# n_iter=10 means it will try 10 random combinations
rdm_search = RandomizedSearchCV(
    estimator=lgbm,
    param_distributions=param_dist,
    n_iter=1,
    scoring='accuracy',
    cv=5,
    verbose=3,
    n_jobs=-1
)

# 4. Fit the search (This handles X_train/y_train splitting internally)
rdm_search.fit(X_train, y_train)

# 5. Get the best model
best_model = rdm_search.best_estimator_

print(f"Best Parameters: {rdm_search.best_params_}")
print(f"Best CV Score: {rdm_search.best_score_:.4f}")

# 6. Evaluate on the actual test set
y_pred = best_model.predict(X_test)
print(classification_report(y_test, y_pred))
```

```
Fitting 5 folds for each of 1 candidates, totalling 5 fits
[CV 2/5] END lambda_l1=1, lambda_l2=5, learning_rate=0.01, max_depth=10, m
in_child_samples=30, n_estimators=100, num_leaves=31;, score=0.914 total t
ime=46.2min
[CV 5/5] END lambda_l1=1, lambda_l2=5, learning_rate=0.01, max_depth=10, m
in_child_samples=30, n_estimators=100, num_leaves=31;, score=0.908 total t
ime=47.3min
[CV 3/5] END lambda_l1=1, lambda_l2=5, learning_rate=0.01, max_depth=10, m
in_child_samples=30, n_estimators=100, num_leaves=31;, score=0.911 total t
ime=47.3min
[CV 4/5] END lambda_l1=1, lambda_l2=5, learning_rate=0.01, max_depth=10, m
in_child_samples=30, n_estimators=100, num_leaves=31;, score=0.913 total t
ime=47.3min
[CV 1/5] END lambda_l1=1, lambda_l2=5, learning_rate=0.01, max_depth=10, m
in_child_samples=30, n_estimators=100, num_leaves=31;, score=0.917 total t
ime=47.3min
Best Parameters: {'num_leaves': 31, 'n_estimators': 100, 'min_child_sample
s': 30, 'max_depth': 10, 'learning_rate': 0.01, 'lambda_l2': 5, 'lambda_l1
': 1}
Best CV Score: 0.9127
              precision    recall  f1-score   support

          A1       0.98      0.93      0.95       210
          A2       0.97      1.00      0.98       288
          A3       0.96      0.79      0.87       348
          A4       0.90      0.98      0.94       561
          A5       0.91      0.98      0.94       531
          B1       0.79      0.82      0.80       350
          B2       0.94      0.98      0.96       398
          B3       0.89      0.98      0.93       567
          B4       0.98      0.82      0.89       491
          B5       0.93      0.99      0.96       529
          C1       0.85      0.83      0.84       418
          C2       0.89      0.85      0.87       394
          C3       0.92      0.95      0.94       298
          C4       0.95      0.96      0.96       240
          C5       0.92      0.97      0.94       230
          D1       0.98      0.93      0.96       183
          D2       0.91      0.91      0.91       263
          D3       0.98      0.96      0.97       229
          D4       0.87      0.94      0.90       192
          D5       0.76      0.96      0.85       170
          E1       0.99      0.81      0.89       150
          E2       1.00      0.68      0.81       128
          E3       0.92      0.90      0.91       108
          E4       0.96      0.82      0.89        90
          E5       0.79      0.91      0.85        82
          F1       0.75      0.55      0.64        65
          F2       0.80      0.76      0.78        49
          F3       0.78      0.81      0.79        36
          F4       0.93      0.85      0.89        33
          F5       0.79      0.83      0.81        23
          G1       0.56      0.45      0.50        20
          G2       0.89      0.50      0.64        16
          G3       0.50      0.11      0.18         9
          G4       0.36      0.36      0.36        11
          G5       0.00      0.00      0.00         6

    accuracy                           0.91      7716
   macro avg       0.84      0.80      0.81      7716
```

```
weighted avg       0.91      0.91      0.91      7716

/home/dantecz/Desktop/Projects/SteppingStone/.venv/lib/python3.8/site-pack
ages/sklearn/metrics/_classification.py:1471: UndefinedMetricWarning: Prec
ision and F-score are ill-defined and being set to 0.0 in labels with no p
redicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/home/dantecz/Desktop/Projects/SteppingStone/.venv/lib/python3.8/site-pack
ages/sklearn/metrics/_classification.py:1471: UndefinedMetricWarning: Prec
ision and F-score are ill-defined and being set to 0.0 in labels with no p
redicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/home/dantecz/Desktop/Projects/SteppingStone/.venv/lib/python3.8/site-pack
ages/sklearn/metrics/_classification.py:1471: UndefinedMetricWarning: Prec
ision and F-score are ill-defined and being set to 0.0 in labels with no p
redicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```

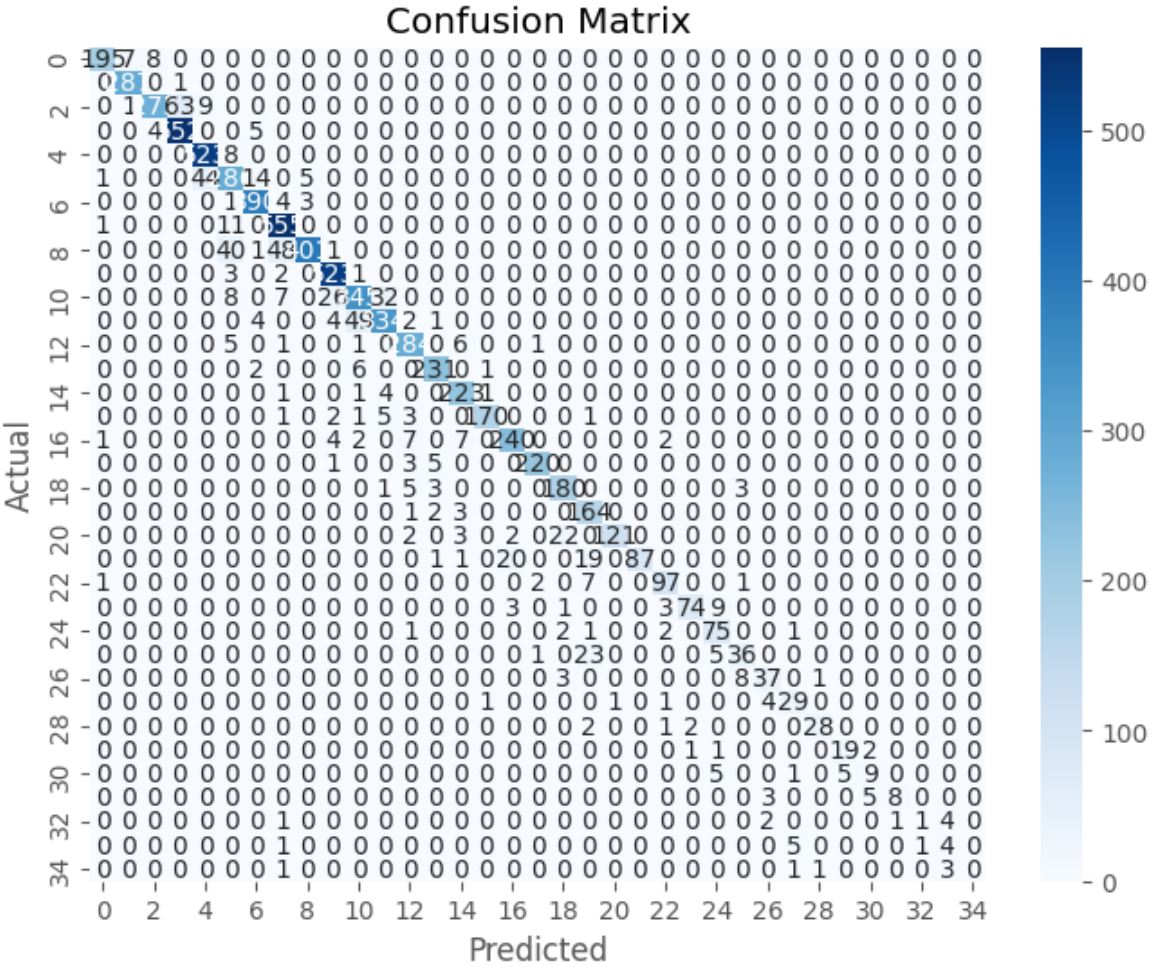
In [ ]:

In [82]:

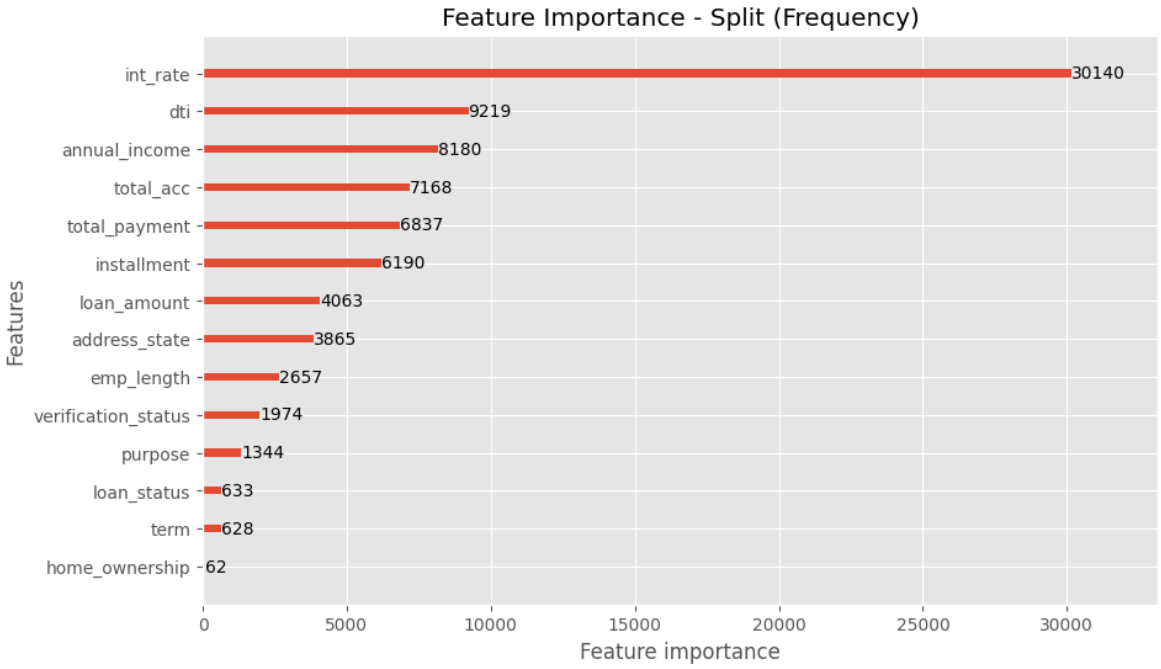
```python
# 4. Evaluation
print("--- Model Performance ---")
print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# 5. Visualizing the Confusion Matrix
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```

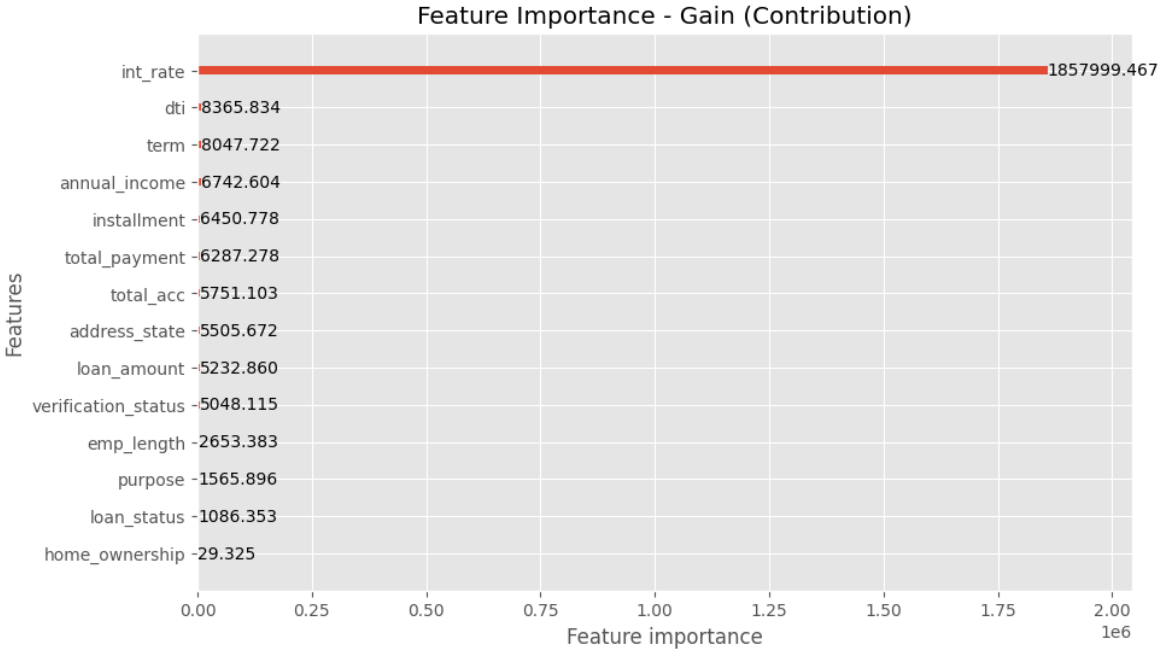
```
--- Model Performance ---
Accuracy: 0.9076

Classification Report:
              precision    recall  f1-score   support

          A1       0.98      0.93      0.95       210
          A2       0.97      1.00      0.98       288
          A3       0.96      0.79      0.87       348
          A4       0.90      0.98      0.94       561
          A5       0.91      0.98      0.94       531
          B1       0.79      0.82      0.80       350
          B2       0.94      0.98      0.96       398
          B3       0.89      0.98      0.93       567
          B4       0.98      0.82      0.89       491
          B5       0.93      0.99      0.96       529
          C1       0.85      0.83      0.84       418
          C2       0.89      0.85      0.87       394
          C3       0.92      0.95      0.94       298
          C4       0.95      0.96      0.96       240
          C5       0.92      0.97      0.94       230
          D1       0.98      0.93      0.96       183
          D2       0.91      0.91      0.91       263
          D3       0.98      0.96      0.97       229
          D4       0.87      0.94      0.90       192
          D5       0.76      0.96      0.85       170
          E1       0.99      0.81      0.89       150
          E2       1.00      0.68      0.81       128
          E3       0.92      0.90      0.91       108
          E4       0.96      0.82      0.89        90
          E5       0.79      0.91      0.85        82
          F1       0.75      0.55      0.64        65
          F2       0.80      0.76      0.78        49
          F3       0.78      0.81      0.79        36
          F4       0.93      0.85      0.89        33
          F5       0.79      0.83      0.81        23
          G1       0.56      0.45      0.50        20
          G2       0.89      0.50      0.64        16
          G3       0.50      0.11      0.18         9
          G4       0.36      0.36      0.36        11
          G5       0.00      0.00      0.00         6

    accuracy                           0.91      7716
   macro avg       0.84      0.80      0.81      7716
weighted avg       0.91      0.91      0.91      7716
```

```
/home/dantecz/Desktop/Projects/SteppingStone/.venv/lib/python3.8/site-pack
ages/sklearn/metrics/_classification.py:1471: UndefinedMetricWarning: Prec
ision and F-score are ill-defined and being set to 0.0 in labels with no p
redicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/home/dantecz/Desktop/Projects/SteppingStone/.venv/lib/python3.8/site-pack
ages/sklearn/metrics/_classification.py:1471: UndefinedMetricWarning: Prec
ision and F-score are ill-defined and being set to 0.0 in labels with no p
redicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
/home/dantecz/Desktop/Projects/SteppingStone/.venv/lib/python3.8/site-pack
ages/sklearn/metrics/_classification.py:1471: UndefinedMetricWarning: Prec
ision and F-score are ill-defined and being set to 0.0 in labels with no p
redicted samples. Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, msg_start, len(result))
```



In [83]:
```
# 1. Importance by 'split' (Frequency)
# This shows how many times a feature was used to make a decision
lgb.plot_importance(model, importance_type='split', max_num_features=20,
plt.title("Feature Importance - Split (Frequency)")
plt.show()

# 2. Importance by 'gain' (Total Information Gain)
# This shows which features actually contributed the most to the model's
lgb.plot_importance(model, importance_type='gain', max_num_features=20, f
plt.title("Feature Importance - Gain (Contribution)")
plt.show()
```





Logistic Regressio, Lasso

I need to first encode the categorical features because they dont work well with logistic regressionm

In [ ]:
```
categorical_features = X_train.select_dtypes(include=['object', 'category
numeric_features = X_train.select_dtypes(include=['number']).columns.toli

print("Categorical Features:", categorical_features)
print("Numeric Features:", numeric_features)
```

```
Categorical Features: ['address_state', 'home_ownership', 'loan_status', '
purpose', 'term', 'verification_status']
Numeric Features: ['emp_length', 'annual_income', 'dti', 'installment', 'i
nt_rate', 'loan_amount', 'total_acc', 'total_payment']
```

In [ ]:
```
from sklearn.compose import ColumnTransformer
```

```
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline

# 2. Create the preprocessing bundles
preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numeric_features),
        ('cat', OneHotEncoder(handle_unknown='ignore', drop='first'), cat
    ])

# 3. Create the final pipeline
# We use drop='first' in OneHotEncoder to avoid the "dummy variable trap"
lasso_pipeline = Pipeline([
    ('preprocessor', preprocessor),
    ('classifier', LogisticRegression(penalty='l1', solver='liblinear', C
])

# 4. Fit the whole thing
lasso_pipeline.fit(X_train, y_train)

# 5. Predict and evaluate
y_pred = lasso_pipeline.predict(X_test)
```

In [ ]:

```
# 4. Evaluation
print("--- Model Performance ---")
print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# 5. Visualizing the Confusion Matrix
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```

```
--- Model Performance ---
Accuracy: 0.2054

Classification Report:
              precision    recall  f1-score   support

          A1       0.61      0.62      0.62       210
          A2       0.61      0.20      0.31       288
          A3       0.23      0.07      0.11       348
          A4       0.35      0.73      0.48       561
          A5       0.37      0.32      0.34       531
          B1       0.22      0.04      0.07       350
          B2       0.18      0.02      0.04       398
          B3       0.19      0.51      0.28       567
          B4       0.15      0.14      0.14       491
          B5       0.14      0.34      0.20       529
          C1       0.12      0.12      0.12       418
          C2       0.09      0.11      0.10       394
          C3       0.14      0.04      0.06       298
          C4       0.11      0.05      0.07       240
          C5       0.13      0.04      0.06       230
          D1       0.10      0.32      0.15       183
          D2       0.05      0.03      0.04       263
          D3       0.09      0.03      0.05       229
          D4       0.05      0.01      0.02       192
          D5       0.08      0.02      0.04       170
          E1       0.03      0.02      0.02       150
          E2       0.10      0.04      0.06       128
          E3       0.16      0.05      0.07       108
          E4       0.17      0.02      0.04        90
          E5       0.24      0.07      0.11        82
          F1       0.14      0.03      0.05        65
          F2       0.00      0.00      0.00        49
          F3       0.50      0.03      0.05        36
          F4       0.00      0.00      0.00        33
          F5       0.00      0.00      0.00        23
          G1       0.00      0.00      0.00        20
          G2       0.00      0.00      0.00        16
          G3       0.00      0.00      0.00         9
          G4       0.00      0.00      0.00        11
          G5       0.00      0.00      0.00         6

    accuracy                           0.21      7716
   macro avg       0.15      0.12      0.11      7716
weighted avg       0.20      0.21      0.17      7716
```

```
/home/dantecz/Desktop/Projects/SteppingStone/.venv/lib/python3.8/site-pack
ages/sklearn/metrics/_classification.py:1471: UndefinedMetricWarning: Prec
ision and F-score are ill-defined and being set to 0.0 in labels with no p
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  _warn_prf(average, modifier, msg_start, len(result))
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  _warn_prf(average, modifier, msg_start, len(result))
```

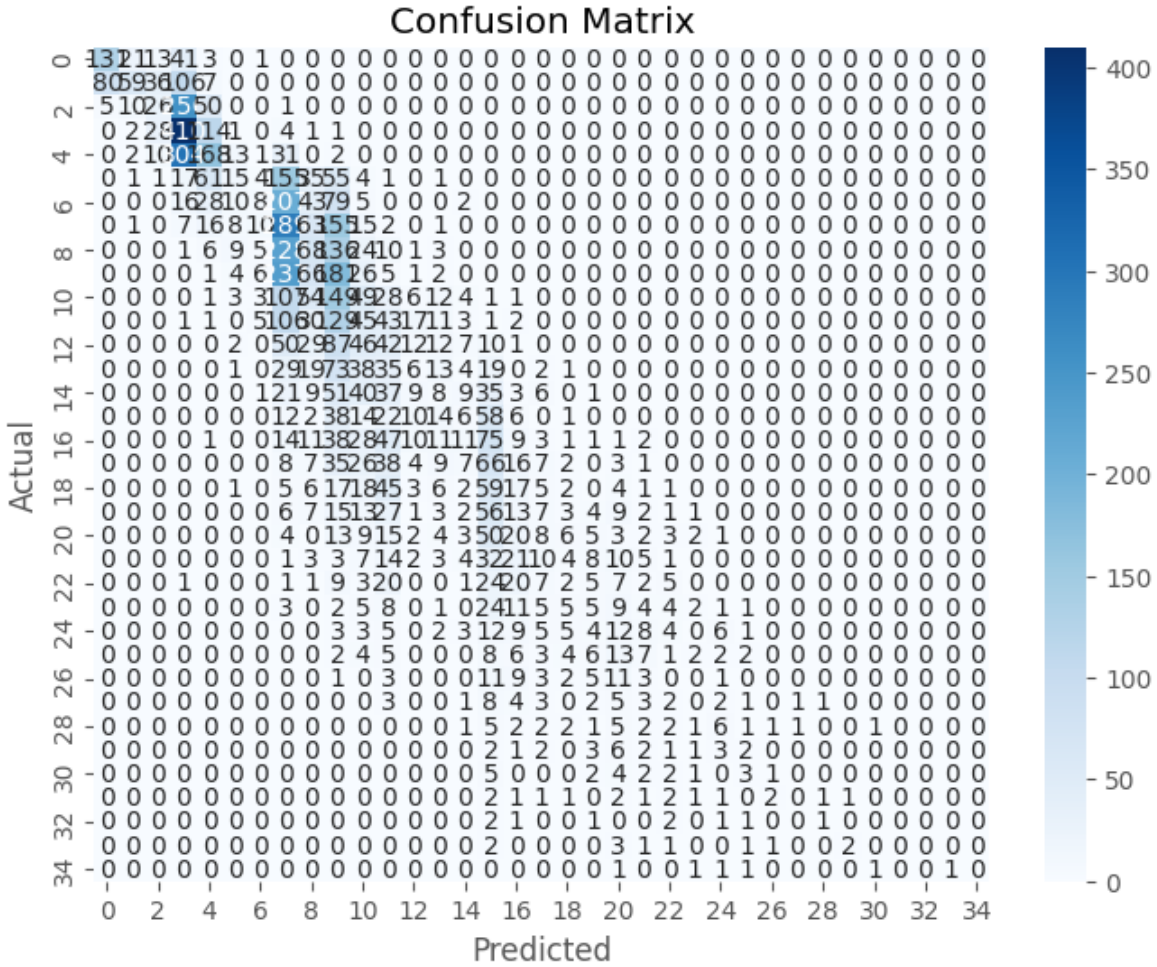

Random Forest

In [ ]:
```
categorical_features
```

Out[ ]:
```
['address_state',
 'home_ownership',
 'loan_status',
 'purpose',
 'term',
 'verification_status']
```

In [ ]:
```
from sklearn.ensemble import RandomForestClassifier
from sklearn.compose import ColumnTransformer
```

```
from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.metrics import classification_report, accuracy_score

# 1. Identify your categorical and numeric columns automatically
categorical_features = X_train.select_dtypes(include=['object', 'category
numeric_features = X_train.select_dtypes(include=['number']).columns.toli

# 2. Create the preprocessor
# Note: Random Forest doesn't technically require scaling, but it doesn't
# However, it MUST have numeric inputs.

dropping = True
if dropping == True:
    preprocessor = ColumnTransformer(
        transformers=[
            ('drop_cols', 'drop', categorical_features), # This explicitl
            # ('cat', OneHotEncoder(handle_unknown='ignore'), categorical
        ], remainder='passthrough') # 'passthrough' keeps numeric columns
if dropping == False:
    preprocessor = ColumnTransformer(
        transformers=[
            # ('drop_cols', 'drop', categorical_features), # This explici
            ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_f
        ], remainder='passthrough') # 'passthrough' keeps numeric columns

# 3. Define the Random Forest Model
rf_model = RandomForestClassifier(
    n_estimators=100,      # Number of trees
    max_depth=None,        # How deep each tree can go
    min_samples_split=2,   # Minimum samples to split a node
    random_state=42,
    n_jobs=-1              # Use all CPU cores for faster training
)

# 4. Create and fit the Pipeline
rf_pipeline = Pipeline([
    ('preprocessor', preprocessor),
    ('classifier', rf_model)
])

rf_pipeline.fit(X_train, y_train)

# 5. Predict and Evaluate
y_pred_rf = rf_pipeline.predict(X_test)

print("--- Random Forest Performance ---")
print(f"Accuracy: {accuracy_score(y_test, y_pred_rf):.4f}")
print(classification_report(y_test, y_pred_rf))
```

```
--- Random Forest Performance ---
Accuracy: 0.6821
              precision    recall  f1-score   support

          A1       0.95      0.92      0.93       210
          A2       0.91      0.91      0.91       288
          A3       0.90      0.68      0.78       348
          A4       0.79      0.86      0.82       561
          A5       0.79      0.93      0.85       531
          B1       0.77      0.65      0.70       350
          B2       0.69      0.75      0.72       398
          B3       0.66      0.85      0.74       567
          B4       0.80      0.67      0.73       491
          B5       0.74      0.77      0.76       529
          C1       0.63      0.68      0.66       418
          C2       0.67      0.69      0.68       394
          C3       0.67      0.56      0.61       298
          C4       0.74      0.52      0.61       240
          C5       0.65      0.60      0.62       230
          D1       0.58      0.60      0.59       183
          D2       0.57      0.69      0.62       263
          D3       0.57      0.69      0.62       229
          D4       0.54      0.51      0.52       192
          D5       0.50      0.51      0.50       170
          E1       0.44      0.49      0.46       150
          E2       0.33      0.31      0.32       128
          E3       0.27      0.30      0.28       108
          E4       0.31      0.26      0.28        90
          E5       0.38      0.35      0.36        82
          F1       0.36      0.25      0.29        65
          F2       0.29      0.08      0.13        49
          F3       0.20      0.08      0.12        36
          F4       0.29      0.15      0.20        33
          F5       0.22      0.09      0.12        23
          G1       0.19      0.15      0.17        20
          G2       0.67      0.12      0.21        16
          G3       0.25      0.11      0.15         9
          G4       0.20      0.09      0.13        11
          G5       0.00      0.00      0.00         6

    accuracy                           0.68      7716
   macro avg       0.53      0.48      0.49      7716
weighted avg       0.68      0.68      0.68      7716

/home/dantecz/Desktop/Projects/SteppingStone/.venv/lib/python3.8/site-pack
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  _warn_prf(average, modifier, msg_start, len(result))
```

In [ ]:
```
# 4. Evaluation
print("--- Model Performance ---")
print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# 5. Visualizing the Confusion Matrix
plt.figure(figsize=(8, 6))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```

```
--- Model Performance ---
Accuracy: 0.2054

Classification Report:
/home/dantecz/Desktop/Projects/SteppingStone/.venv/lib/python3.8/site-pack
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  _warn_prf(average, modifier, msg_start, len(result))
```

```
              precision    recall  f1-score   support

          A1       0.61      0.62      0.62       210
          A2       0.61      0.20      0.31       288
          A3       0.23      0.07      0.11       348
          A4       0.35      0.73      0.48       561
          A5       0.37      0.32      0.34       531
          B1       0.22      0.04      0.07       350
          B2       0.18      0.02      0.04       398
          B3       0.19      0.51      0.28       567
          B4       0.15      0.14      0.14       491
          B5       0.14      0.34      0.20       529
          C1       0.12      0.12      0.12       418
          C2       0.09      0.11      0.10       394
          C3       0.14      0.04      0.06       298
          C4       0.11      0.05      0.07       240
          C5       0.13      0.04      0.06       230
          D1       0.10      0.32      0.15       183
          D2       0.05      0.03      0.04       263
          D3       0.09      0.03      0.05       229
          D4       0.05      0.01      0.02       192
          D5       0.08      0.02      0.04       170
          E1       0.03      0.02      0.02       150
          E2       0.10      0.04      0.06       128
          E3       0.16      0.05      0.07       108
          E4       0.17      0.02      0.04        90
          E5       0.24      0.07      0.11        82
          F1       0.14      0.03      0.05        65
          F2       0.00      0.00      0.00        49
          F3       0.50      0.03      0.05        36
          F4       0.00      0.00      0.00        33
          F5       0.00      0.00      0.00        23
          G1       0.00      0.00      0.00        20
          G2       0.00      0.00      0.00        16
          G3       0.00      0.00      0.00         9
          G4       0.00      0.00      0.00        11
          G5       0.00      0.00      0.00         6

    accuracy                           0.21      7716
   macro avg       0.15      0.12      0.11      7716
weighted avg       0.20      0.21      0.17      7716

/home/dantecz/Desktop/Projects/SteppingStone/.venv/lib/python3.8/site-pack
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  _warn_prf(average, modifier, msg_start, len(result))
```
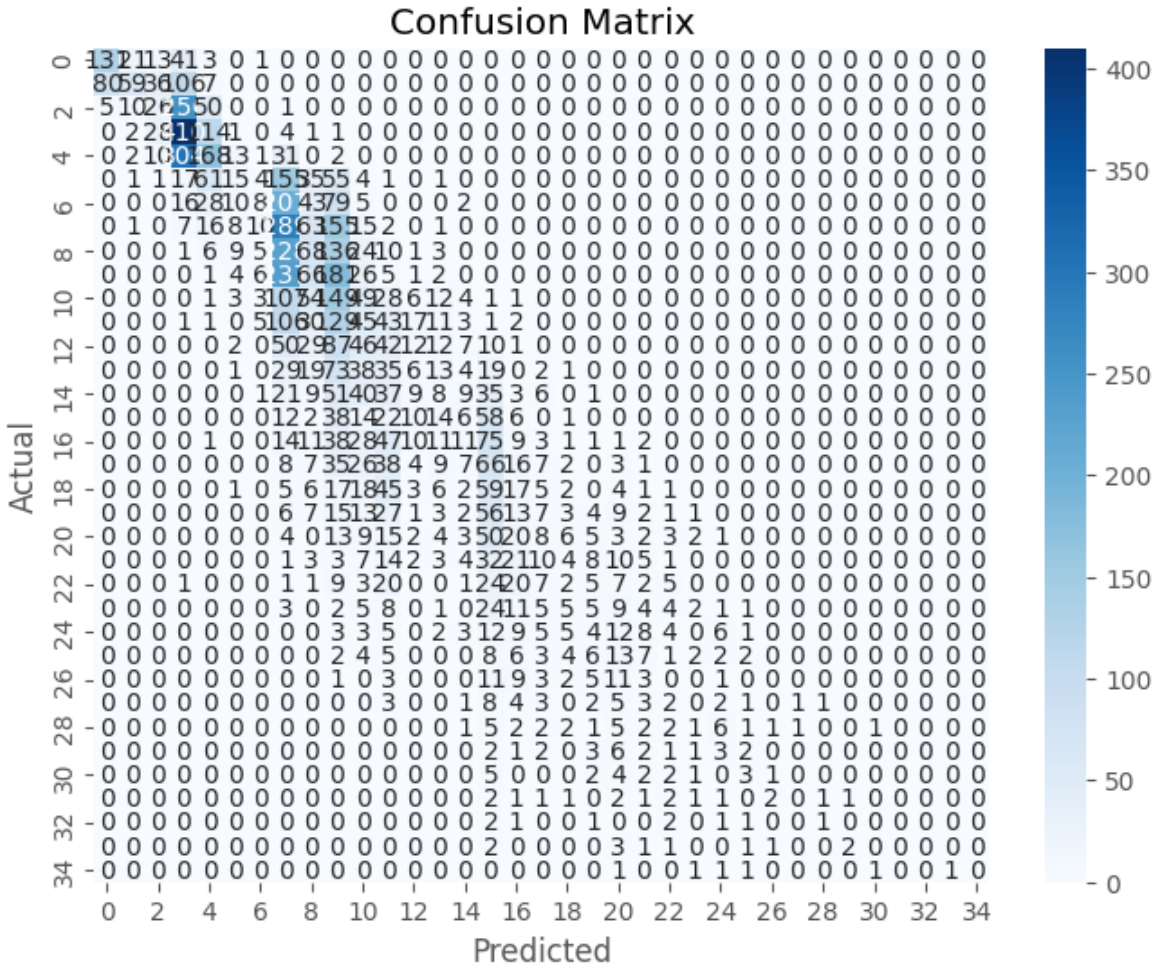
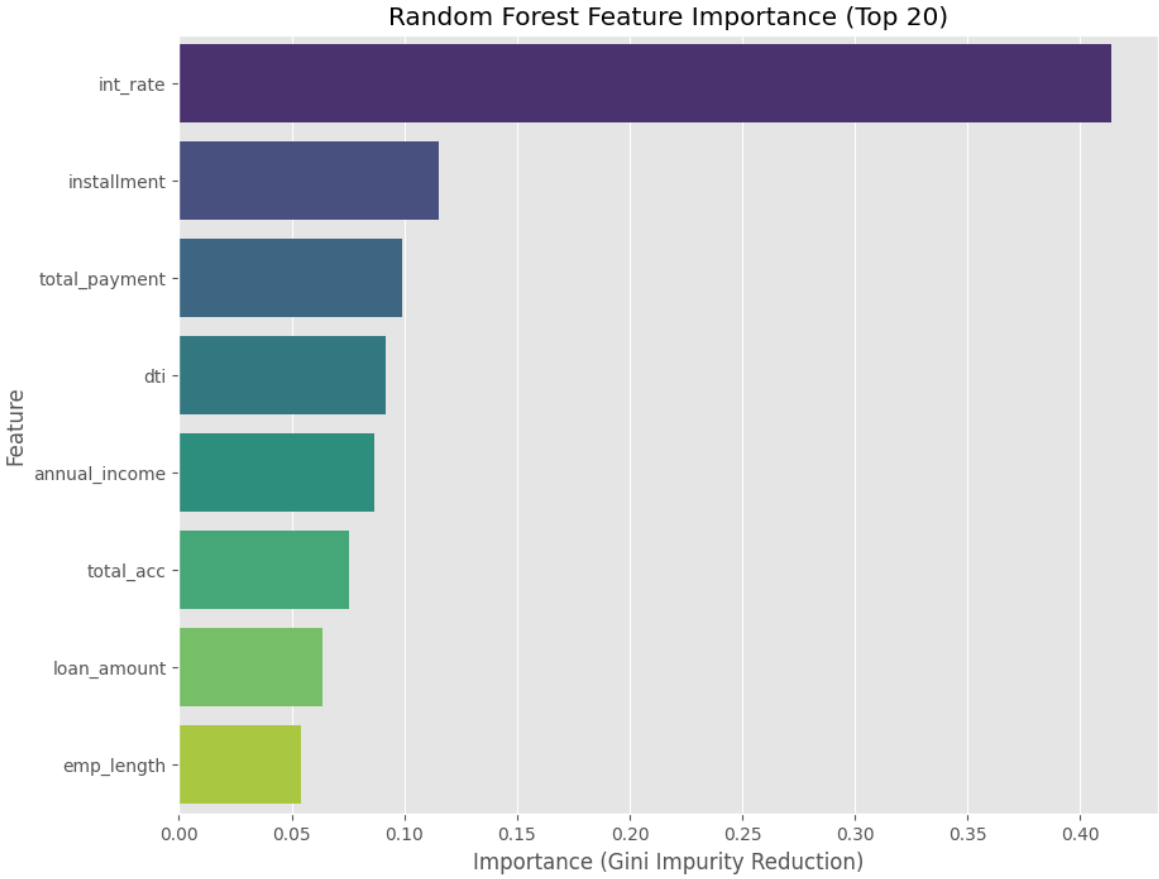



In [ ]:
```
# 1. Get the feature names from the One-Hot Encoder
# This looks at the 'cat' step of the preprocessor inside the pipeline
if dropping == True:
    ohe_feature_names = []
else:
    ohe_feature_names = rf_pipeline.named_steps['preprocessor']\
                        .transformers_[0][1]\
                        .get_feature_names_out(categorical_features)

# 2. Combine with the numeric feature names (which didn't change)
all_feature_names = list(ohe_feature_names) + numeric_features

# 3. Grab the importance values from the classifier
importances = rf_pipeline.named_steps['classifier'].feature_importances_

# 4. Create a DataFrame for easy plotting
feat_imp_df = pd.DataFrame({
    'feature': all_feature_names,
    'importance': importances
}).sort_values(by='importance', ascending=False)

# 5. Plot the top 20
plt.figure(figsize=(10, 8))
sns.barplot(data=feat_imp_df.head(20), x='importance', y='feature', palet
plt.title('Random Forest Feature Importance (Top 20)')
plt.xlabel('Importance (Gini Impurity Reduction)')
plt.ylabel('Feature')
plt.show()
```

```
/tmp/ipykernel_67783/2347287647.py:24: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be remove
d in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for
the same effect.

  sns.barplot(data=feat_imp_df.head(20), x='importance', y='feature', pale
tte='viridis')
```



In [ ]:
```
feat_imp_df
```

Out[ ]:

	feature	importance
4	int_rate	0.414034
3	installment	0.115155
7	total_payment	0.098791
2	dti	0.092012
1	annual_income	0.086528
6	total_acc	0.075762
5	loan_amount	0.063799
0	emp_length	0.053920

In [ ]: